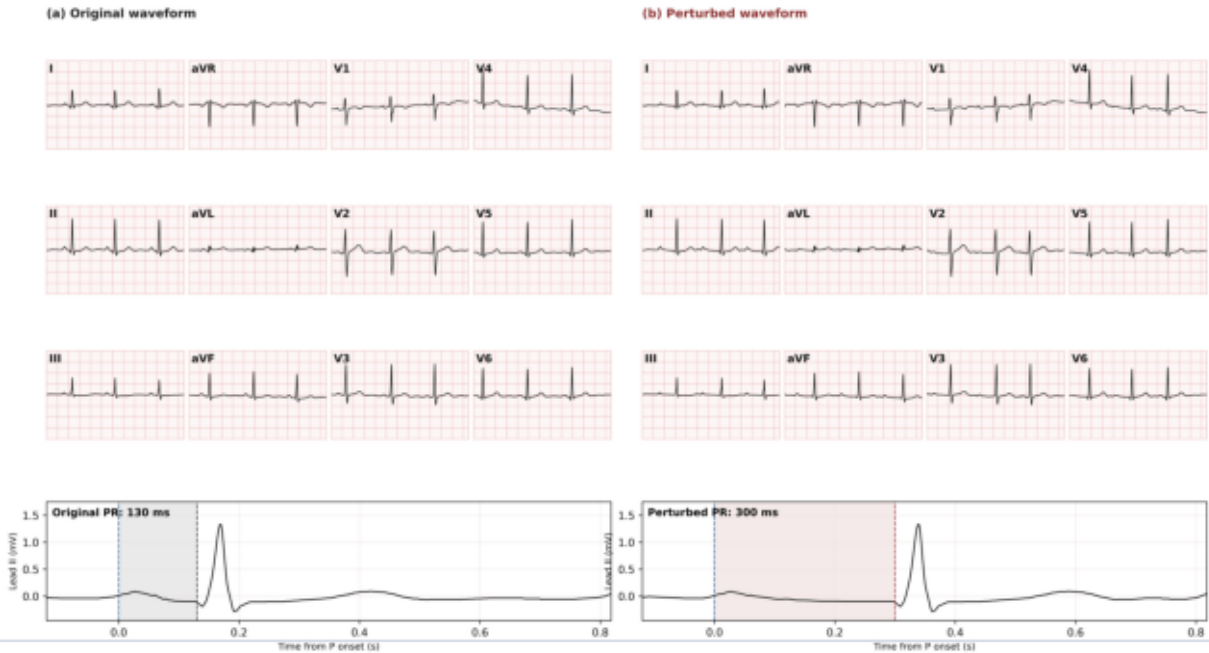


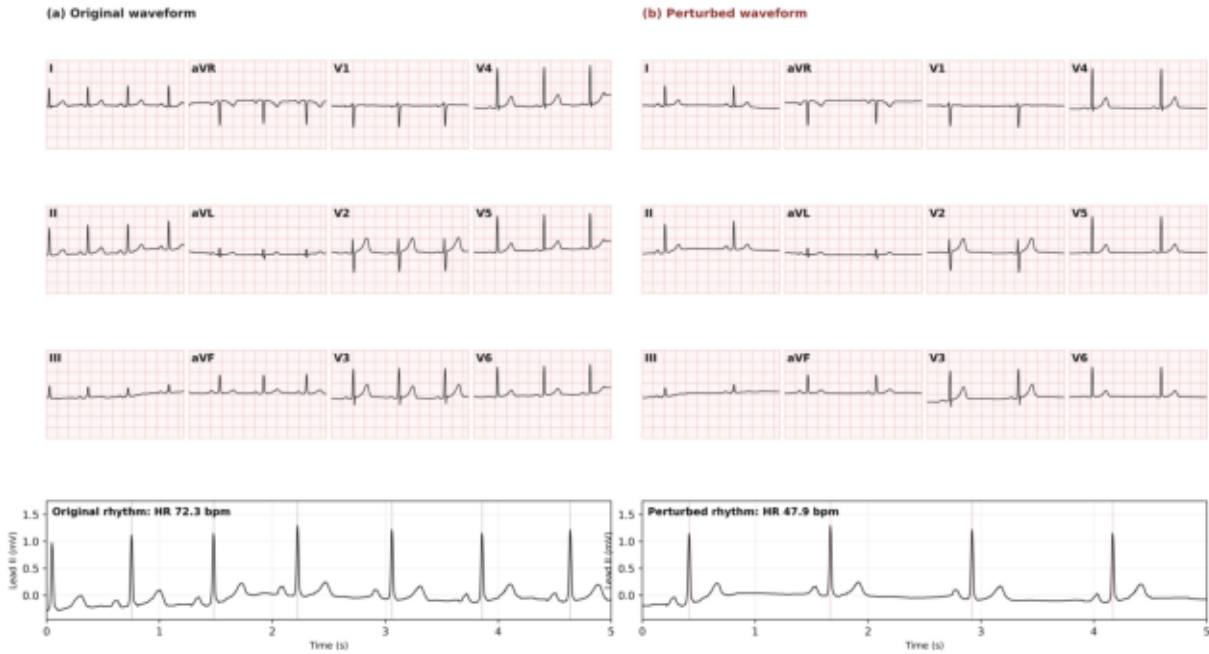
Case 1 — PR-interval prolongation

Waveform source: PTB-XL - train split



Case 2 — Heart-rate reduction

Waveform source: PTB-XL - train split



Case 1 · Morphology-preserving PR prolongation

PR 130 → 298 ms · HR 74.6 bpm preserved

P and QRS-T copied exactly; PQ extended and TP shortened within each RR cycle.

Synthetic measurement MCQ

- A. HR 74.6 · PR 130 · QRS 96 · QTc 393.7 → no 1° AV block
- B. HR 48.0 · PR 298 · QRS 96 · QTc 393.7 → bradycardia + 1° AV block
- ✓ C. HR 74.6 · PR 298 · QRS 96 · QTc 393.7 → first-degree AV block
- D. HR 74.6 · PR 298 · QRS 130 · QTc 393.7 → 1° AV block + IVCD

Original A → Counterfactual C

Qwen3.8 re-measured normal rate, prolonged PR and narrow QRS; excluded A/B/D.

Qwen3.8-max thinking: PASS · boxed C · 8,845 assistant characters

signal edit → re-measure → MCQ → multimodal CoT → strict boxed-answer gate

Case 2 · Morphology-preserving sinus-rate slowing

HR 72.3 → 47.9 bpm · PR/QRS preserved

P-QRS-T copied exactly; only the post-T/pre-P isoelectric segment was extended.

Synthetic measurement MCQ

- A. HR 72.3 · PR 150 · QRS 48 · QTc 348.5 → normal rate
- ✓ B. HR 47.9 · PR 150 · QRS 48 · QTc 348.5 → sinus bradycardia
- C. HR 47.9 · PR 220 · QRS 48 · QTc 348.5 → bradycardia + 1° AV block
- D. HR 47.9 · PR 150 · QRS 48 · QTc 490 → bradycardia + long QT

Original A → Counterfactual B

Qwen3.8 measured RR ≈1.25 s, normal PR/QRS/QTc and selected sinus bradycardia.

Qwen3.8-max thinking: PASS · boxed B · 5,934 assistant characters

signal edit → re-measure → MCQ → multimodal CoT → strict boxed-answer gate